

CRUSHIT

Gamified Pedal Can Crusher · Urban Recycling Reimagined

IMechE CADathon 2026 · SDG 11: Sustainable Cities & Communities



THE PROBLEM

Aluminium cans: valuable, recyclable and constantly dumped in the wrong bin

~2.5x

more cans
fit when they are crushed

75%

of aluminium ever made
is still in active use today

95%

energy saved recycling
aluminium compared to smelting it raw

Root cause: Recycling bins are inconvenient, passive, and boring. People skip them especially at events like football matches where waste peaks. **CrushIt makes the right choice the fun choice.**

OUR SOLUTION

CrushIt: A pedal-powered, gamified aluminium can crusher bin



Zero electricity

Entirely foot-pedal driven. No battery, no grid connection needed.



Compact crushing

Reduces can volume by approx. 60%. 2.5× more cans per collection cycle.



Gamified engagement

Dual-sided crusher: pick Team A or B to answer trivia/vote. Count tracked live. Fun for all ages.



Safety-first design

Manual lid closure before pedal activates. Finger entrapment possibility eliminated.



Liquid waste capture

Grid lower platform drains residual liquid to sealed tray. Thus, clean crushes and no mess.



STADIUM CRUSHIT

Messi **vs** **Ronaldo**

Who gets more cans? Pick a side!

Side A
505
cans

Side B
470
cans

Crush your can. Cast your vote.

CAD & ENGINEERING DESIGN



1 Foot Pedal Actuation

Pedal drives a 2-hinge toggle linkage, converting foot force to vertical platform travel.

2 Upper Fixed Platform

Spike on punctures can top on contact to releases pressure & drain residual liquid before full crush, mitigating explosion caused by carbonation.

3 Lower Moving Platform

Grid-pattern base travels upward to crush the can. Open grid allows liquid to drain into sealed collection tray below.

4 Slanted Floor + Exit Slot

Floor slightly inclined toward a side slot at base. After crushing, the can slides on to the collection tray.

5 Safety Lid + Pedal Lock

User manually closes lid before use. Mechanical interlock prevents pedal from actuating until lid is fully shut.

MATERIALS & COST VIABILITY

Component	Material	Justification	Est. Unit Cost
Crushing platforms	304 Stainless Steel	Corrosion-resistant, food-safe, high compressive strength	£15 – 25
Housing enclosure	HDPE (injection moulded)	Weatherproof, lightweight, recyclable; low cost for long life cycle.	£8 – 20
Hinge linkage	Galvanised mild steel	High fatigue life for high-cycle repetitive loading	£5 – 10
Liquid tray	Polypropylene (PP)	Chemical-resistant, food-grade, easy to injection mould	£3 – 6
Puncture spike	Hardened stainless steel	Puncture resistance, hygiene compliance, corrosion-free	£2 – 4

Manufacturing at Scale

HDPE injection moulding – enclosure ~£0.80–2.00/unit at volume

Sheet metal stamping – platforms: precise & repeatable

Estimated total unit cost: **£55–100 at scale**

SDG 11 Alignment

- ✓ Cuts bin contamination and landfill waste
- ✓ 2.5× bin capacity → fewer collection vehicles
- ✓ Passive design — zero ongoing energy use
- ✓ All materials recyclable at end-of-life

FUTURE DEVELOPMENT & IMPACT



Finger-Safety IR Sensor

IR sensor detects hands inside crush zone. Auto-locks pedal until clear of hands. More Convenient.



Live Digital Display

LED panel shows crush count, bin fill %, and live game scores. Low power draw. Drives competition and repeat engagement.



Vending Machine Pairing

Pair CrushIt units adjacent to beverage vending machines. Deposit tokens dispensed per crush for discounts.



Fill-Level Monitoring

Smart sensors report bin fill to council waste teams. Eliminates unnecessary collections, optimising routes.

CrushIt — making recycling something people actually want to do.